COMPLETE LASER CHARACTERIZATION REPORT

Spatial, temporal, spectral, and power characterization

Coherent COMPASS 315M-100 (532 nm CW) and Thorlabs GSL106A (1064 nm)

Measurement domains	Spatial 532 nm; temporal 1064 nm; spectral 532/1064 nm; optical power
Primary instruments	Basler acA3088-57uc; PDA36A + DSO6104A; iHR550/Synapse Plus; PM100D/S121C
Source corpus	119 archive entries; 36 native RAW frames; 22 supplemental PNGs; 19 oscilloscope CSVs; 40 HORIBA spectra
Analysis status	All supplied files catalogued; complete machine-readable outputs and executed notebooks included
Report date	2026-08-30

Graduate-level analysis with explicit confidence and limitation statements

Original archive SHA-256: c58900255d515fab38bb8883aa2cf632de8d63608b10f0956d7ba76f622d03f2

Abstract

A complete characterization was performed using all supplied spatial, temporal, and spectral acquisitions for a Coherent COMPASS 315M-100 532 nm continuous-wave laser and a Thorlabs GSL106A 1064 nm source operated in CW, picosecond, nominal 1 ns, and nominal 65 ns modes. The source archive contains 119 catalogued entries. Native BayerRG12 camera data were used for 532 nm beam-profile analysis; oscilloscope records were analyzed for repetition rate and electrical response; HORIBA spectra were analyzed for center wavelength, bandwidth, sampling adequacy, slit dependence, and transform-limit bounds; and PM100D/S121C power measurements were converted to pulse-energy bounds or ranges at 200 kHz.

The central conclusion is not that every nominal source specification was directly measured. Rather, the dataset supports a hierarchy of claims. Sensor-plane beam dimensions, selected-lobe geometry, the trigger-chain repetition rate, electrical waveform widths, average optical power, pulse energy, and broadband picosecond spectral width are supported. A final standards-grade M2, picosecond or 1 ns optical pulse duration, intrinsic narrow linewidth, intrinsic timing jitter, peak power, and traceable absolute wavelength near 1064 nm are not supported by the present apparatus or calibration. The report preserves this distinction throughout.

Primary result

The data are scientifically useful when each result is named precisely. Electrical FWHM is not optical pulse duration; measured narrow-line FWHM is not intrinsic linewidth; and a provisional caustic fit is not a certified M2 measurement.

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Executive summary

Table 1. Executive confidence summary

Domain	Supported result	Critical qualification
Spatial 532 nm	Selected-lobe dimensions and repeatability are usable; provisional M2 = 2.539 and 1.814 along principal axes.	Provisional M2 only; saturation, no dedicated dark, reflections, and limited valid caustic coverage.
Temporal 1064 nm	Repetition rate is approximately 200.0003 kHz. Electrical FWHM values: ps 20.8896 ns, 1 ns 32.3680 ns, 65 ns 49.0250 ns.	Widths describe the detector/electronics response, not the ps or 1 ns optical pulses.
Power / energy	ps >6.75 pJ; 1 ns >47.0 pJ; 65 ns 1.65-1.70 nJ per pulse. CW powers: 532 >130 mW; 1064 >16 mW.	Bounds/range preserved. Peak power is unsupported without optical duration and pulse shape.
532 nm spectra	Peak positions near 532 nm are meaningful at instrument level.	Narrow FWHM values are undersampled and instrument limited; not intrinsic linewidth.
1064 nm spectra	Narrow spectra show strong slit broadening; ps spectral FWHM is 8.492-9.606 nm.	Absolute 1064 nm center is unverified because calibration used only 435.8/546.1 nm Hg lines.
ps transform limit	0.170-0.192 ps for Gaussian, 0.122-0.137 ps for sech^2 assumptions.	Conditional lower bounds, not measured durations.

1. Experimental system and data corpus

1.1 Optical sources

Table 2. Laser sources

Role	Model	Modes	Power/operation note
532 nm source	Coherent COMPASS 315M-100	CW	Reported >130 mW
1064 nm source	Thorlabs GSL106A	CW, ps, 1 ns, 65 ns	Pulsed modes at 200 kHz

Both sources were reportedly warmed for approximately two minutes before measurement. That duration is documented rather than assumed to represent thermal steady state. Any observed drift can therefore include warm-up behavior as well as measurement noise and source fluctuations.

1.2 Spatial measurement chain

The 532 nm beam directly intercepted a Basler acA3088-57uc CMOS camera. No camera lens, relay, or magnification optics were present. The optical sequence was laser, reflective ND filter(s), one mirror, then camera. The camera used BayerRG12 data with auto gain, auto exposure, and color

correction disabled. Exposures were approximately 8-16 us. The highest-intensity lobe was defined as the intended beam when reflections or secondary spots were present.

For focused measurements, z was recorded in centimeters from an approximately 100 mm Thorlabs plano-convex lens, with positive z farther from the lens. The no-lens series used an arbitrary zero. Independent frames at each position were retained as repeats. No dedicated dark frame was acquired.

1.3 Temporal measurement chain

The GSL106A was externally triggered by an Agilent 33120A function generator. A BNC T connected the function-generator output to the GSL106A trigger input and oscilloscope CH3; CH3 therefore monitors the electrical drive/trigger. The detector signal was recorded on CH1 from a legacy Thorlabs PDA36A set to 0 dB gain. The oscilloscope was a DSO6104A operated with DC coupling. BNC cables were shorter than 1 m. No separate external feed-through termination was used; files labeled 50 ohm are interpreted as using the oscilloscope input termination.

Reflective ND filters attenuated the beam for the 1 ns and 65 ns detector measurements. A mechanical chopper was present at 987 Hz for demonstration but did not intercept or modulate the analyzed beam. The chopper frequency is therefore not expected in the laser pulse train.

1.4 Spectral measurement chain

Spectra were acquired in free space through the front entrance of a HORIBA iHR550 with Synapse Plus CCD, using 300, 1200, and 1800 groove/mm gratings. LabSpec 6 used signal-dark processing, binning 1, one accumulation, and real-time-display acquisitions where 0 s denotes live operation. Slits were fully closed or at the minimum practical separation unless a filename records a larger value.

Each grating was checked with mercury lines at 435.8 nm and 546.1 nm by applying wavelength offsets until the two visible peaks matched to four significant figures. No traceable near-IR reference was available. Consequently, the 532 nm region is interpolated near the calibration span, whereas approximately 1064 nm is a long extrapolation.

1.5 Power measurement chain

Optical power was measured with a Thorlabs PM100D console and S121C sensor. GSL106A measurements were made in free space immediately after the F280APC-1064 fiber collimator and before any optical attenuation. The user-recorded symbols are retained: “>” is a strict lower bound and “~” is approximate. The 65 ns observation was approximately 333 uW, with a recalled range of 330-340 uW.

1.6 Data corpus and completeness

Table 3. Source archive inventory by domain

Domain assignment	Files
Spatial (532 nm)	58
Spectral (532/1064 nm)	42

Domain assignment	Files
Temporal (1064 nm)	19
Total archive entries	119

The original archive SHA-256 is c58900255d515fab38bb8883aa2cf632de8d63608b10f0956d7ba76f622d03f2. Every non-directory archive entry is listed in Appendix A and in the machine-readable source_data_inventory.csv. The analysis retains both native and supplemental images, all oscilloscope exports, all HORIBA spectra, and supporting acquisition files. No supplied acquisition was intentionally removed from the project record.

2. Analysis principles and reporting conventions

2.1 Measured, derived, conditional, and unsupported quantities

- Measured: extracted directly from calibrated instrument axes or image intensity data, with stated processing.
- Derived: calculated from measured values by a defined equation, such as pulse energy from average power and repetition rate.
- Provisional: a numerical fit exists, but one or more validity requirements are not satisfied.
- Conditional: valid only under an explicit model, such as transform-limited Gaussian or sech^2 pulse shape.
- Unsupported: the measurement chain does not resolve the requested physical quantity, so no numerical claim is made.

2.2 Uncertainty policy

No arbitrary uncertainty or confidence interval is introduced. Repeatability is reported where independent acquisitions exist. Fit covariance is used only as a local numerical diagnostic and is not promoted to a full experimental uncertainty. Systematic terms that were not measured - for example z-position uncertainty, power-sensor calibration certificate, exact slit width, wavelength-reference uncertainty near 1064 nm, or detector impulse response - remain explicit limitations.

2.3 Spatial processing

Native 12-bit Bayer data are preferred over exported 8-bit PNGs. The two green samples in each 2 x 2 Bayer cell are combined without interpolation, giving a green-superpixel pitch of 4.8 um from the 2.4 um physical pixel pitch. A robust planar background estimate is used because no matched dark frame exists. The highest-intensity connected lobe defines the primary beam. Second-moment D4sigma diameters, centroid, principal-axis angle, ellipticity, saturation fraction, clipping indicators, and secondary-structure diagnostics are computed. Gaussian fits are used for localization/residual diagnostics, not as proof of Gaussian beam quality.

2.4 Temporal processing

Oscilloscope metadata and waveform samples are parsed from the native CSV exports. Long records are folded at the trigger repetition period. Hysteretic edge detection prevents ringing from being counted as extra triggers. Electrical FWHM, rise/fall behavior, equivalent width, overshoot, undershoot, ringing, pulse-to-pulse variation, repetition rate, and timing residuals are calculated. The detector response is not deconvolved into optical duration because the response is non-Gaussian and insufficiently calibrated for that purpose.

2.5 Spectral processing

HORIBA text files are parsed with their metadata, including decimal-comma cases. Baseline is estimated from spectral edges. Peak wavelength, centroid, nonparametric FWHM, RMS width, equivalent width, frequency-domain bandwidth, resolving power, and samples per FWHM are calculated. Fits are rejected or downgraded when undersampling or instrument limitation dominates. Narrow-line widths are not quadratically deconvolved from an assumed catalog resolution; a measured instrument line-spread function would be required.

3. Spatial characterization of the 532 nm COMPASS beam

3.1 Data quality and beam morphology

The native spatial set contains 36 BayerRG12 frames at 12 axial configurations, with three independent acquisitions per configuration. The beam is frequently accompanied by secondary spots, rings, or reflections. Fifteen of the 36 native frames contain saturated pixels, and most saturation occurs in the focused waist region. The selected highest-intensity lobe is therefore the physically intended beam by experimental definition, but the full field is not a clean single Gaussian mode.

3.2 Second-moment beam dimensions and repeatability

The most defensible spatial outputs are the per-frame and per-position selected-lobe centroid, principal-axis D4sigma diameters, ellipticity, and orientation. These are sensor-plane measurements because the camera directly intercepted the beam. Repeated frames provide Type A repeatability at each recorded z position. The exact numerical rows are retained in the spatial output tables and the executed spatial notebook.

> **Interpretation**
>
> A repeatable sensor-plane diameter can be valid even when a global Gaussian model is poor. Conversely, a visually plausible Gaussian fit does not remove saturation or secondary-lobe bias from second moments.

3.3 Provisional caustic and M2

A numerical caustic fit to the selected-lobe principal axes gives provisional M2 values of 2.539 on the major axis and 1.814 on the minor axis, with fitted waist radii of approximately 209 um and 172 um

and fitted waist locations of approximately 10.34 cm and 11.28 cm from the lens reference. The differing axes are consistent with a non-circular or astigmatic beam/measurement state.

Why M2 is not final

The waist region is heavily saturated; only a small subset of focused positions is strictly unsaturated; secondary structure affects moments; no dedicated dark exists; z-position uncertainty was not recorded; and the usable caustic span is insufficient in fitted Rayleigh-range units. Report the fit as provisional, not as a standards-grade laser specification.

3.4 Spatial conclusion

The COMPASS beam can be characterized as non-circular at the camera plane, with position-dependent selected-lobe dimensions and measurable repeatability. The present dataset is adequate for qualitative propagation behavior and provisional caustic modeling. It is not adequate for a final ISO-style M2 claim. The reported >130 mW CW power is independent of the camera analysis and should not be combined with saturated pixel values to infer irradiance.

4. Temporal and power characterization of the 1064 nm GSL106A

4.1 Repetition rate

Table 4. Repetition-rate results

Operating mode	Long-record trigger-chain repetition rate
1 ns	200000.295 Hz
65 ns	200000.370 Hz
ps	200000.384 Hz

All three long records agree with 200 kHz. The approximately 0.3-0.4 Hz offsets from exactly 200000 Hz are small at the record level and may include timebase, edge-selection, and finite-record effects. The result is best described as the repetition rate of the complete function-generator/trigger/laser/detector/scope measurement chain.

4.2 Electrical pulse-response widths

Table 5. Electrical response widths

Mode	Measured electrical FWHM	Physical interpretation
ps	20.8896 ns	Detector/electronics limited; not a picosecond optical duration
1 ns	32.3680 ns	Detector/electronics limited; not a 1 ns optical duration

Mode	Measured electrical FWHM	Physical interpretation
65 ns	49.0250 ns	Partially resolved but detector-convolved and ringing-sensitive

The observed bipolar/ringing response is incompatible with a simple Gaussian instrument response. Therefore, subtracting a nominal detector width in quadrature would produce a falsely precise optical duration. The ps and 1 ns optical pulses are unresolved. The nominal 65 ns mode is closer to the detector response scale, but its measured 49.0250 ns electrical FWHM still cannot be substituted for the source optical pulse duration.

4.3 Optical power and pulse energy

Table 6. Power and energy results

Source/mode	Average power	Pulse energy	Photon scale
532 CW	>130 mW	-	>3.482e+17 photons/s
1064 CW	>16 mW	-	>8.570e+16 photons/s
1064 ps, 200 kHz	>1.35 uW	>6.75 pJ	>3.616e+07 photons/pulse
1064 1 ns, 200 kHz	>9.4 uW	>47.0 pJ	>2.517e+08 photons/pulse
1064 65 ns, 200 kHz	~333 uW; 330-340 uW	1.65-1.70 nJ	8.838e+09-9.106e+09 photons/pulse

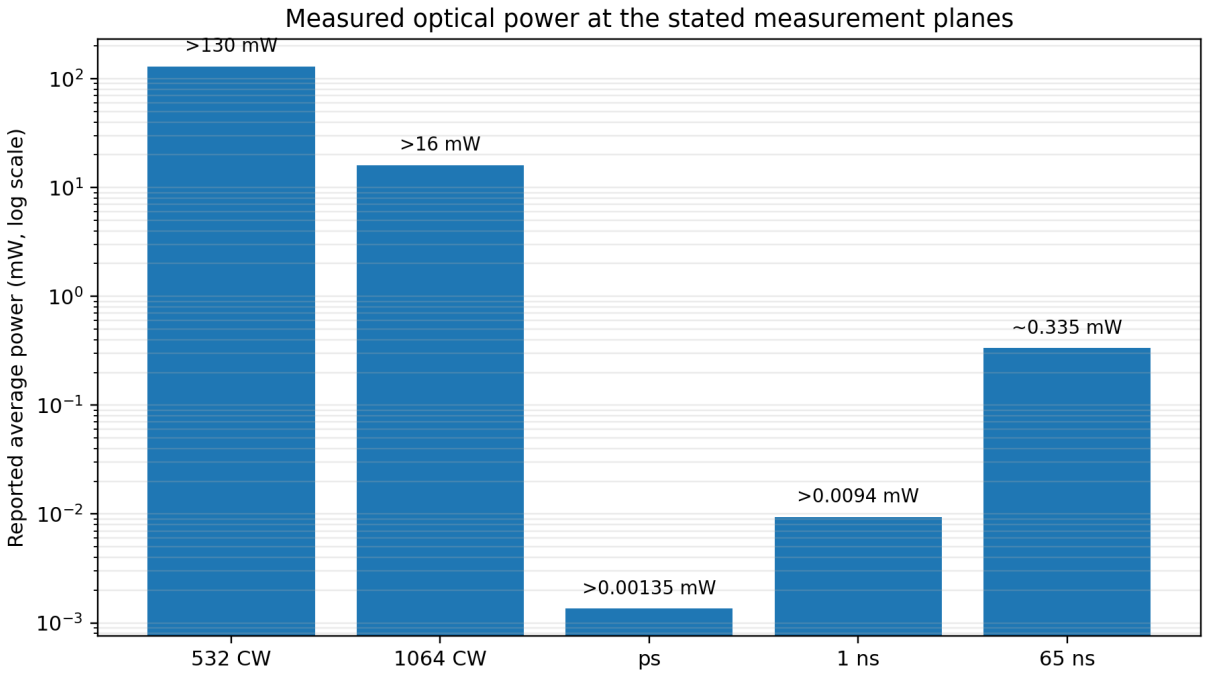


Figure 6. Reported optical powers. The log axis spans the CW and pulsed average-power regimes; lower-bound and approximate semantics are shown in the labels.

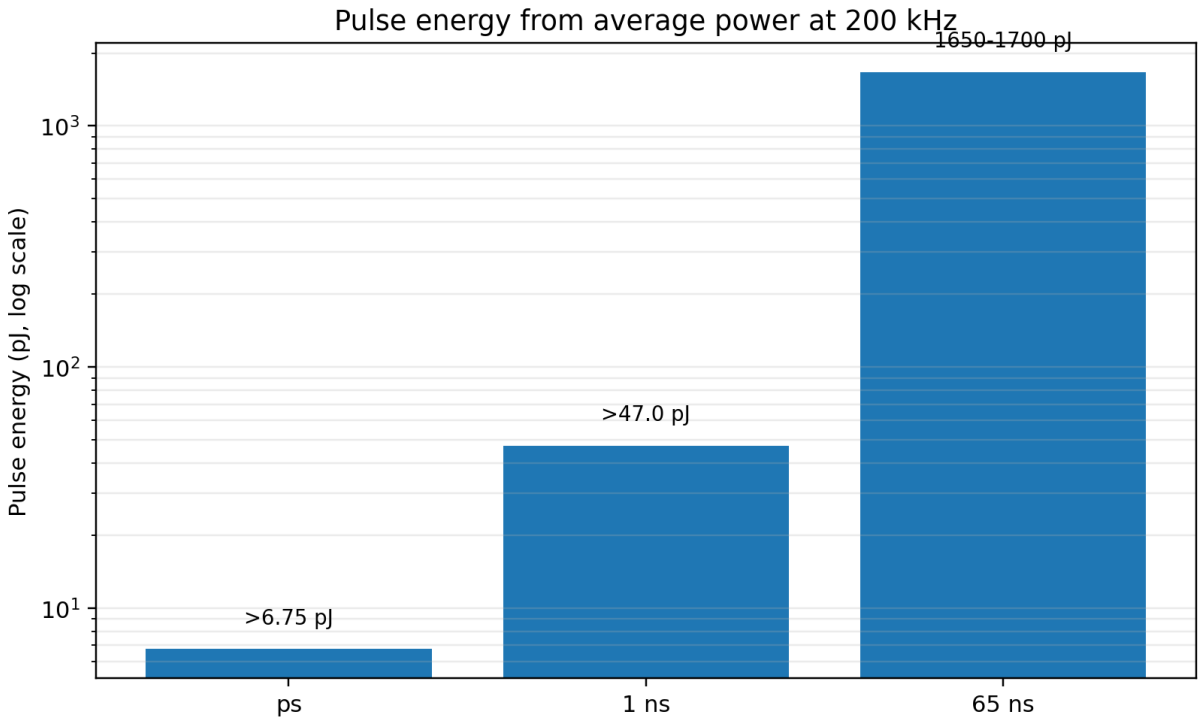


Figure 7. Pulse energy derived from average power at 200 kHz. This calculation does not require a pulse-shape assumption.

Peak power is not a measured result

Using a nominal duration would produce a conditional average power during the pulse, not a calibrated peak power. The report therefore stops at pulse energy unless an autocorrelator, fast enough calibrated detector, streak camera, FROG, SPIDER, or equivalent optical timing measurement is added.

4.4 Temporal stability and timing interpretation

Pulse-to-pulse amplitude, area, and trigger-to-response timing residuals in the output tables describe system-level repeatability. They include contributions from the 33120A, BNC T and cables, GSL106A trigger response, PDA36A, oscilloscope timebase/sampling, threshold selection, and channel delay. They must not be labeled intrinsic laser jitter without an independently characterized reference and measurement chain.

5. Spectral characterization at 532 nm and 1064 nm

5.1 Calibration and sampling

The 435.8 and 546.1 nm Hg references substantially constrain the green spectral region, but they do not establish traceable accuracy at 1064 nm. An offset correction can remove a local zero error; it does not independently verify grating dispersion, spectrometer model, order effects, or wavelength-

dependent systematic error over a 500 nm extrapolation. Accordingly, absolute 1064 nm centers are reported as instrument readings with low absolute confidence.

Width versus center

A nearly constant wavelength offset shifts a spectrum without strongly changing FWHM. Thus the broad ps spectral width is more defensible than the absolute ps center wavelength, while narrow linewidths remain limited by slit width, instrument response, and sampling.

5.2 532 nm CW spectra

Across the 300, 1200, and 1800 groove/mm acquisitions, measured FWHM values span approximately 0.0363-0.320 nm. The narrow peaks contain only about 1.58-2.15 sampled points per FWHM. This is below what is needed for a robust line-shape or intrinsic-linewidth determination. The dependence on grating and sampling is itself evidence that the measured width is controlled by the instrument configuration.

5.3 Narrow 1064 nm spectra and slit series

In the 1200 groove/mm CW slit series, the measured FWHM increases from approximately 0.0559 nm at the nominal 10 um setting to approximately 0.2246 nm at 200 um. This approximately fourfold change with slit width demonstrates instrumental broadening. No intrinsic GSL106A linewidth can be extracted without measuring the instrument line-spread function under the same grating, slit, wavelength, detector, and acquisition conditions.

The narrow peaks around 1059-1061 nm are physically plausible within a broad nominal 1064 nm source band, but their absolute accuracy is unverified. The correct scientific statement is not that the spectrometer is necessarily wrong; it is that the visible-only calibration does not provide enough evidence to decide whether the offset is source-related or instrument-related.

5.4 Picosecond-mode spectra

Table 7. Picosecond spectral characterization

Quantity	Result	Interpretation
Spectral FWHM	8.492-9.606 nm	Direct, well-sampled broadband result
Frequency FWHM	2.297-2.591 THz	Derived from wavelength axis
Gaussian transform limit	0.170-0.192 ps	Conditional lower bound, zero spectral phase
sech^2 transform limit	0.122-0.137 ps	Conditional lower bound, zero spectral phase

The ps spectra are the strongest spectral result in the dataset because tens to hundreds of samples span the FWHM. The transform-limit calculations are useful physical lower bounds. They are not measured pulse widths: chirp or arbitrary spectral phase can lengthen the pulse substantially while leaving the spectrum unchanged.

6. Cross-domain interpretation

6.1 Consistency of temporal and spectral evidence

The ps optical spectrum supports a sub-picosecond transform-limited lower bound, while the PDA36A/scope trace is approximately 20.9 ns wide. These statements are not contradictory. They show that the optical pulse is far shorter than the detector-chain response. The temporal trace confirms triggering and optical detection at 200 kHz, whereas the spectrum constrains the minimum possible pulse duration under ideal phase assumptions.

6.2 Power-energy-mode scaling

At a common 200 kHz repetition rate, the 65 ns mode carries approximately 35 times the pulse energy of the 1 ns lower bound and approximately 244-252 times the ps lower bound. These ratios compare the reported measurement levels, not intrinsic source conversion efficiencies, because two values are lower bounds and operating conditions may differ beyond the nominal mode selection.

6.3 Thermal stabilization and session dependence

A roughly two-minute warm-up may be insufficient to establish long-term thermal equilibrium for the lasers and spectrometer. The data therefore characterize the recorded session. Repeating the experiment after a documented stabilization interval would separate short-term acquisition repeatability from warm-up drift and long-term reproducibility.

6.4 Optical artifacts and measurement planes

Spatial reflections originate in a path containing reflective ND filters and one mirror; they should not automatically be attributed to the source transverse mode. Power was measured at the GSL106A collimator output before attenuation, while temporal detection used attenuation for 1 ns and 65 ns. These are different measurement planes and must not be combined to infer detector responsivity or absolute detector power without the ND transmission and geometry.

7. Uncertainty, confidence, and validity assessment

Table 8 assigns confidence to each named quantity. A result can be high confidence in one sense and unsuitable for another. For example, 32.3680 ns is a well-determined electrical FWHM but not a valid estimate of a 1 ns optical pulse duration.

Table 8. Measurement-confidence matrix

Quantity	Result	Confidence	Basis / limitation
532 nm optical power	>130 mW	Moderate	Direct PM100D/S121C reading reported as a strict lower bound; no calibration certificate or repeat series supplied.
532 nm sensor-plane centroid and beam dimensions	Per-frame values in spatial tables	Moderate	Direct camera data and independent repeats; limited by saturation, no dedicated dark frame, background structure, and multi-spot reflections.

Quantity	Result	Confidence	Basis / limitation
532 nm M2, major axis	2.539 provisional	Low / provisional	Caustic fit exists, but saturation near waist, limited valid z coverage, unknown z uncertainty, and secondary structure prevent a standards-grade result.
532 nm M2, minor axis	1.814 provisional	Low / provisional	Same limitations as major axis.
1064 nm repetition rate	Approximately 200.0003 kHz	High for measurement chain	Long records and hysteretic trigger-edge detection agree across modes; this is trigger-chain repetition rate, not a standalone intrinsic-jitter certification.
ps electrical response FWHM	20.8896 ns	High as electrical width	The exported waveform supports this electrical response width; it is not the optical pulse duration.
1 ns electrical response FWHM	32.3680 ns	High as electrical width	The detector/electronics do not resolve a 1 ns optical pulse.
65 ns electrical response FWHM	49.0250 ns	Moderate as electrical width	Partially resolved but detector-convolved and affected by ringing; not a calibrated optical duration.
ps pulse energy	>6.75 pJ	Moderate lower bound	Derived exactly from the reported lower-bound average power and measured/nominal 200 kHz rate.
1 ns pulse energy	>47.0 pJ	Moderate lower bound	Derived exactly from the reported lower-bound average power and 200 kHz rate.
65 ns pulse energy	1.65-1.70 nJ	Moderate approximate range	Derived from the reported 330-340 uW range at 200 kHz.
Peak power in any pulsed mode	Not determined	Not supported	Requires trustworthy optical pulse shape and duration; detector traces are insufficient.
532 nm intrinsic linewidth	Not determined	Not supported	Narrow features are undersampled and instrument limited.
1064 nm narrow spectral center	Approximately 1059-1061 nm measured	Low absolute confidence	No near-IR reference; visible offset calibration is extrapolated far beyond 546.1 nm.
1064 nm narrow intrinsic linewidth	Not determined	Not supported	Strong slit dependence and insufficient sampling demonstrate instrument limitation.
ps spectral FWHM	8.492-9.606 nm	Moderate to high	Broad feature is well sampled; width is more robust than absolute center wavelength.
ps transform-limit lower bound	0.170-0.192 ps Gaussian; 0.122-0.137 ps sech^2	Conditional	Calculated from spectral intensity width assuming a pulse shape and zero spectral phase; not a measured temporal duration.

Quantity	Result	Confidence	Basis / limitation
Absolute wavelength accuracy near 1064 nm	Unverified	Not established	Requires a traceable near-IR line/reference measured with the same grating and slit configuration.

7.1 Dominant spatial limitations

- Saturation near the focused waist
- No matched camera dark frame
- Secondary spots/reflections and aperture sensitivity
- Unknown z-position uncertainty and approximate 100 mm lens focal length
- Insufficient strictly valid caustic coverage for final M2

7.2 Dominant temporal limitations

- Legacy PDA36A impulse response and ringing
- No independent detector-response calibration at the exact gain/load condition
- No optical autocorrelation or ultrafast temporal diagnostic
- System-level rather than source-only jitter measurement
- No detector responsivity/ND transmission chain used for absolute instantaneous optical power

7.3 Dominant spectral limitations

- Visible-only wavelength references for measurements near 1064 nm
- Unknown exact slit widths when “closed/minimum”
- Narrow peaks sampled by roughly two points per FWHM
- No measured instrument line-spread function at 1064 nm
- One accumulation and approximately two-minute system warm-up

7.4 Dominant power limitations

- Inequality readings provide bounds rather than equality estimates
- No measurement-specific calibration certificate or repeated reading series supplied
- No uncertainty contribution for beam centering, sensor uniformity, or environmental conditions
- Peak power requires optical duration and pulse-shape information that is not available

8. Conclusions and recommended follow-up measurements

8.1 Final conclusions

- The complete dataset supports quantitative sensor-plane spatial characterization of the selected 532 nm primary lobe, but the global M2 values 2.539 and 1.814 remain provisional.
- The GSL106A trigger/measurement chain operates at approximately 200.0003 kHz in ps, 1 ns, and 65 ns modes.
- Electrical detector widths of 20.8896 ns, 32.3680 ns, and 49.0250 ns are reproducible waveform metrics, not direct optical pulse durations.

- Reported powers imply pulse energies >6.75 pJ, >47.0 pJ, and 1.65-1.70 nJ for ps, 1 ns, and 65 ns modes at 200 kHz.
- Narrow 532 nm and 1064 nm linewidths are instrument/sampling limited. The broad ps spectral FWHM of 8.492-9.606 nm is a defensible broadband result.
- The ps spectrum implies conditional transform-limit lower bounds of 0.170-0.192 ps (Gaussian) or 0.122-0.137 ps (sech^2), not measured durations.
- Absolute wavelength near 1064 nm remains unverified because the calibration references stop at 546.1 nm.

8.2 Highest-value follow-up measurements

Table 9. Recommended follow-up work

Objective	Recommended measurement
Final M2	Repeat focused caustic with no saturation, dedicated dark/background, recorded z uncertainty, more positions extending several Rayleigh ranges on both sides, and controlled removal of ghost reflections.
ps/1 ns duration	Use an autocorrelator/FROG/SPIDER or a detector and sampling chain with independently verified bandwidth appropriate to the pulse.
65 ns duration	Measure detector impulse response at the actual PDA36A gain/load and use a monotonic, non-ringing high-bandwidth detector for direct pulse-shape validation.
Intrinsic linewidth	Measure a narrow reference line and instrument line-spread function at the same wavelength, grating, slit, detector, and acquisition settings.
Absolute 1064 nm wavelength	Use a traceable near-IR reference, wavemeter, or calibrated line near 1064 nm.
Power uncertainty	Retain PM100D/S121C calibration certificate, wavelength correction, zeroing/background, repeated readings, beam centering, and environmental records.
Stability	Document warm-up and acquire time series after thermal stabilization for source and spectrometer.

Appendix A. Complete source-file audit

This appendix lists every non-directory entry in 532 and 1064.zip. The machine-readable version includes CRC32, compressed size, and any matching SHA-256 from the retained source manifest. Total entries: 119.

Table A1. Complete source-file inventory

ID	Domain	Archive path	Bytes	CRC32
1	Spectral (532/1064 nm)	532 and 1064/1064nm_Spectrum/Laser_1064nm_Mode_CW_RTD_0s_Slit_0um_Grad_300_ND_3.txt	18,654	d74cfc15
2	Temporal	532 and 1064/Pulse_Data/Improper_Impedance_CH1_1Mohm/1_ns_Mode/Mode_1ns_ND_0_Gain_0db_ExtTrigger_200kHz_Chopper_987Hz_TimeDiv_108us.csv	76,649,935	0db875ff
3	Temporal	532 and 1064/Pulse_Data/Improper_Impedance_CH1_1Mohm/1_ns_Mode/Mode_1ns_ND_0_Gain_0db_ExtTrigger_200kHz_Chopper_987Hz_TimeDiv_2us.csv	1,418,368	de86beae
4	Temporal	532 and 1064/Pulse_Data/Improper_Impedance_CH1_1Mohm/1_ns_Mode/Mode_1ns_ND_0_Gain_0db_ExtTrigger_200kHz_Chopper_987Hz_TimeDiv_300ns.csv	217,238	60e5cb75
5	Temporal	532 and 1064/Pulse_Data/Improper_Impedance_CH1_1Mohm/1_ns_Mode/Mode_1ns_ND_0_Gain_0db_ExtTrigger_200kHz_Chopper_987Hz_TimeDiv_33ns.csv	75,230	fffc5d96
6	Temporal	532 and 1064/Pulse_Data/Improper_Impedance_CH1_1Mohm/65_ns_Mode/Mode_65ns_ND_1_Gain_0db_ExtTrigger_200kHz_Chopper_987Hz_TimeDiv_104us.csv	73,700,316	1e299d6e
7	Temporal	532 and 1064/Pulse_Data/Improper_Impedance_CH1_1Mohm/65_ns_Mode/Mode_65ns_ND_1_Gain_0db_ExtTrigger_200kHz_Chopper_987Hz_TimeDiv_2us.csv	1,420,896	b30cf17c
8	Temporal	532 and 1064/Pulse_Data/Improper_Impedance_CH1_1Mohm/65_ns_Mode/Mode_65ns_ND_1_Gain_0db_ExtTrigger_200kHz_Chopper_987Hz_TimeDiv_66ns.csv	72,665	3978f32b

ID	Domain	Archive path	Bytes	CRC32
9	Temporal	532 and 1064/Pulse_Data/Improper_Impedance_CH1_1Mohm/Picosecond_Mode/Mode_ps_ND_0_Gain_0db_ExtTrigger_200kHz_Chopper_987Hz_TimeDiv_104us.csv	68,718,055	c3e4b89a
10	Temporal	532 and 1064/Pulse_Data/Improper_Impedance_CH1_1Mohm/Picosecond_Mode/Mode_ps_ND_0_Gain_0db_ExtTrigger_200kHz_Chopper_987Hz_TimeDiv_2.5us.csv	1,636,749	f10440da
11	Temporal	532 and 1064/Pulse_Data/Improper_Impedance_CH1_1Mohm/Picosecond_Mode/Mode_ps_ND_0_Gain_0db_ExtTrigger_200kHz_Chopper_987Hz_TimeDiv_66us.csv	67,265	be68f198
12	Temporal	532 and 1064/Pulse_Data/Proper_Impedance_CH1_50ohm/1_ns_Mode/Mode_1ns_ND_0_Gain_0db_ExtTrigger_200kHz_TimeDiv_100ns.csv	210,224	ef5f9ef3
13	Temporal	532 and 1064/Pulse_Data/Proper_Impedance_CH1_50ohm/1_ns_Mode/Mode_1ns_ND_0_Gain_0db_ExtTrigger_200kHz_TimeDiv_20us.csv	39,993,335	8dcc8914
14	Temporal	532 and 1064/Pulse_Data/Proper_Impedance_CH1_50ohm/1_ns_Mode/Mode_1ns_ND_0_Gain_0db_ExtTrigger_200kHz_TimeDiv_2us.csv	4,003,505	1076f258
15	Temporal	532 and 1064/Pulse_Data/Proper_Impedance_CH1_50ohm/65_ns_Mode/Mode_65ns_ND_1_Gain_0db_ExtTrigger_200kHz_TimeDiv_2us.csv	4,073,712	fc6ee8a4
16	Temporal	532 and 1064/Pulse_Data/Proper_Impedance_CH1_50ohm/65_ns_Mode/Mode_65ns_ND_1_Gain_0db_ExtTrigger_200kHz_TimeDiv_50us.csv	104,161,618	9c218554
17	Temporal	532 and 1064/Pulse_Data/Proper_Impedance_CH1_50ohm/65_ns_Mode/Mode_65ns_ND_1_Gain_0db_ExtTrigger_200kHz_TimeDiv_66ns.csv	140,939	a27c11ed
18	Temporal	532 and 1064/Pulse_Data/Proper_Impedance_CH1_50ohm/Picosecond_Mode/Mode_ps_ND_0_Gain_0db_ExtTrigger_200kHz_TimeDiv_100ns.csv	212,524	a5427bc3
19	Temporal	532 and 1064/Pulse_Data/Proper_Impedance_CH1_50ohm/Picosecond_Mode/Mode_ps_ND_0_Ga	4,130,624	46bcdbe3

ID	Domain	Archive path	Bytes	CRC32
		in_0db_ExtTrigger_200kHz_TimeDiv_2us.csv		
20	Temporal	532 and 1064/Pulse_Data/Proper_Impedance_CH1_50ohm/Picosecond_Mode/Mode_ps_ND_0_Gain_0db_ExtTrigger_200kHz_TimeDiv_50us.csv	103,225,679	3fc11c82
21	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/532nm_Spectrum/Laser_532nm_CW_Acq_0_5s_Slit_0um_Grad_1200.txt	17,789	c5eae835
22	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/532nm_Spectrum/Laser_532nm_CW_Acq_0_5s_Slit_0um_Grad_1800.txt	17,785	a2b841ba
23	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/532nm_Spectrum/Laser_532nm_CW_Acq_0_5s_Slit_0um_Grad_300.txt	17,899	921e434c
24	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/532nm_Spectrum/Laser_532nm_CW_Acq_1s_Slit_0um_Grad_300.txt	18,089	32d4e987
25	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_CW_Acq_0_5s_Slit_100um_Grad_1200.txt	17,764	5f6baa0a
26	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_CW_Acq_0_5s_Slit_10um_Grad_1200.txt	17,822	983854ed
27	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_CW_Acq_0_5s_Slit_10um_Grad_300.txt	18,823	45e950cc
28	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_CW_Acq_0_5s_Slit_10um_Grad_300_(1).txt	18,816	07f2f7c1
29	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_CW_Acq_0_5s_Slit_10um_Grad_300_(2).txt	18,779	533d56ce
30	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_CW_Acq_0_5s_Slit_10um_Grad_	18,781	af7dc3e5

ID	Domain	Archive path	Bytes	CRC32
		300_(3).txt		
31	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_CW_Acq_0_5s_Slit_200um_Grad_1200.txt	17,787	58fec0dd
32	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_CW_Acq_0_5s_Slit_50um_Grad_1200.txt	17,790	587ca489
33	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_65nanoseconds_Acq_0_5s_Slit_10um_Grad_1200.txt	18,250	0aa026fd
34	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_65nanoseconds_Acq_0_5s_Slit_10um_Grad_1200_(1).txt	18,336	354a60ad
35	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_65nanoseconds_Acq_0_5s_Slit_10um_Grad_300 (2).txt	17,819	36d461f9
36	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_65nanoseconds_Acq_0_5s_Slit_10um_Grad_300.txt	18,479	160b13a6
37	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_65nanoseconds_Acq_0_5s_Slit_10um_Grad_300_(1).txt	18,606	99c902ad
38	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_65nanoseconds_Acq_0_5s_Slit_10um_Grad_300_(2).txt	18,617	14824924
39	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_65nanoseconds_Acq_0_5s_Slit_10um_Grad_300_(3).txt	18,491	10822508
40	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_65nanoseconds_Acq_0_5s_Slit_10um_Grad_300_(4).txt	17,826	4c9d09d3
41	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_65nanoseconds_Acq_0_5s_Slit_10um_Grad_300_(5).txt	17,793	1f15defe

ID	Domain	Archive path	Bytes	CRC32
42	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_65nanoseconds_Acq_0_5s_Slit_10um_Grad_300_(6).txt	17,832	b06c1f35
43	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_65nanoseconds_Acq_0_5s_Slit_50um_Grad_300.txt	18,556	0588e100
44	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_65nanoseconds_Acq_0_5s_Slit_50um_Grad_300_(1).txt	18,451	f32617e3
45	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_65nanoseconds_Acq_1s_Slit_100um_Grad_300.txt	18,259	1b42252c
46	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_65nanoseconds_Acq_1s_Slit_100um_Grad_300_(1).txt	18,395	2f6002d3
47	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_65nanoseconds_Acq_1s_Slit_50um_Grad_300.txt	18,291	fa6cb01d
48	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_65nanoseconds_Acq_1s_Slit_50um_Grad_300_(1).txt	18,321	79025b20
49	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_65nanoseonds_RTD_0s_Slit_10um_Grad_1200.txt	18,043	23d42c3f
50	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_CW_RTD_0s_Slit_0um_Grad_300_ND_3.txt	18,654	d74cfc15
51	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_picoseconds_Acq_0_5s_Slit_10um_Grad_1200.txt	17,767	71b5fd8d
52	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_picoseconds_Acq_0_5s_Slit_10um_Grad_1200_(1).txt	17,753	959b5af0
53	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Las	18,392	8f3b6d50

ID	Domain	Archive path	Bytes	CRC32
		er_1064nm_Mode_picoseconds_Acq_0_5s_Slit_10um_Grad_300.txt		
54	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_picoseconds_Acq_0_5s_Slit_10um_Grad_300_(1).txt	18,118	aacb2f6f
55	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_picoseconds_Acq_0_5s_Slit_10um_Grad_300_(2).txt	17,999	782ec488
56	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_picoseconds_Acq_0_5s_Slit_10um_Grad_300_(3).txt	18,381	b3671a5f
57	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_picoseconds_Acq_0_5s_Slit_10um_Grad_300_(4).txt	18,407	258bc56e
58	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_picoseconds_Acq_0_5s_Slit_10um_Grad_300_(5).txt	18,375	412352e8
59	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_picoseconds_Acq_0_5s_Slit_50um_Grad_300_(3).txt	17,968	ebf68731
60	Spectral (532/1064 nm)	532 and 1064/Spectral_Data/1064nm_Spectrum/Laser_1064nm_Mode_picoseconds_Acq_1s_Slit_50um_Grad_300.txt	17,806	1f2b2700
61	Spatial	532 and 1064/532nm_NoFocal_Gain0_Exposure10k_BayerRG12/Background_Reference.png	8,443,624	71f74cff
62	Spatial	532 and 1064/532nm_NoFocal_Gain0_Exposure10k_BayerRG12/Position_0mm.png	8,881,449	b650823a
63	Spatial	532 and 1064/532nm_NoFocal_Gain0_Exposure10k_BayerRG12/Position_100mm.png	8,508,048	c3f53a59
64	Spatial	532 and 1064/532nm_NoFocal_Gain0_Exposure10k_BayerRG12/Position_10mm.png	9,403,602	36d224ee
65	Spatial	532 and 1064/532nm_NoFocal_Gain0_Exposure10k_BayerRG12/Position_150mm.png	8,545,265	af95ad74

ID	Domain	Archive path	Bytes	CRC32
66	Spatial	532 and 1064/532nm_NoFocal_Gain0_Exposure10k_BayerRG12/Position_200mm.png	8,309,769	df0a39bf
67	Spatial	532 and 1064/532nm_NoFocal_Gain0_Exposure10k_BayerRG12/Position_20mm.png	9,503,732	df4865d1
68	Spatial	532 and 1064/532nm_NoFocal_Gain0_Exposure10k_BayerRG12/Position_250mm.png	9,236,402	512f4a87
69	Spatial	532 and 1064/532nm_NoFocal_Gain0_Exposure10k_BayerRG12/Position_30mm.png	9,333,595	3c8aa8cf
70	Spatial	532 and 1064/532nm_NoFocal_Gain0_Exposure10k_BayerRG12/Position_40mm.png	8,581,120	0398ef22
71	Spatial	532 and 1064/532nm_NoFocal_Gain0_Exposure10k_BayerRG12/Position_50mm.png	8,535,617	9b620d37
72	Spatial	532 and 1064/532nm_NoFocal_Gain0_Exposure10k_BayerRG12/Position_60mm.png	8,534,960	32296c3d
73	Spatial	532 and 1064/532nm_NoFocal_Gain0_Exposure10k_BayerRG12/Position_70mm.png	8,482,287	4bbaefa5
74	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_0cm_ND_4_7_BayerRG12_Focal_No_1.raw	12,747,264	8247554d
75	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_0cm_ND_4_7_BayerRG12_Focal_No_2.raw	12,747,264	2df18fee
76	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_0cm_ND_4_7_BayerRG12_NoFocal_0.raw	12,747,264	a070c1f0
77	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_10cm_ND_4_7_BayerRG12_Focal_No_0.raw	12,747,264	dddfeb85
78	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_10cm_ND_4_7_BayerRG12_Focal_No_1.raw	12,747,264	e3612821

ID	Domain	Archive path	Bytes	CRC32
79	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_10cm_ND_4_7_BayerRG12_Focal_No_2.raw	12,747,264	fbcbdfa9
80	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_10cm_ND_4_7_BayerRG12_Focal_Yes_0.raw	12,747,264	73e9682b
81	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_10cm_ND_4_7_BayerRG12_Focal_Yes_1.raw	12,747,264	59eea79b
82	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_10cm_ND_4_7_BayerRG12_Focal_Yes_2.raw	12,747,264	4fb5bf60
83	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_11cm_ND_4_7_BayerRG12_Focal_Yes_0.raw	12,747,264	61895873
84	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_11cm_ND_4_7_BayerRG12_Focal_Yes_1.raw	12,747,264	984dd410
85	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_11cm_ND_4_7_BayerRG12_Focal_Yes_2.raw	12,747,264	8df35458
86	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_12cm_ND_4_7_BayerRG12_Focal_Yes_0.raw	12,747,264	295adb7e
87	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_12cm_ND_4_7_BayerRG12_Focal_Yes_1.raw	12,747,264	d2cf2ead
88	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_12cm_ND_4_7_BayerRG12_Focal_Yes_2.raw	12,747,264	c1ed624a
89	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_15cm_ND_4_7_BayerRG12_Focal_Yes_0.raw	12,747,264	c64056fa
90	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_P	12,747,264	a1e0914d

ID	Domain	Archive path	Bytes	CRC32
		osition_15cm_ND_4_7_BayerRG12_Focal_Yes_1.raw		
91	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_15cm_ND_4_7_BayerRG12_Focal_Yes_2.raw	12,747,264	84479c1b
92	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_20cm_ND_4_7_BayerRG12_Focal_No_0.raw	12,747,264	8bf2cc1c
93	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_20cm_ND_4_7_BayerRG12_Focal_No_1.raw	12,747,264	570150ae
94	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_20cm_ND_4_7_BayerRG12_Focal_No_2.raw	12,747,264	9b741918
95	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_20cm_ND_4_7_BayerRG12_Focal_Yes_0.raw	12,747,264	522d7792
96	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_20cm_ND_4_7_BayerRG12_Focal_Yes_1.raw	12,747,264	a4231ae6
97	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_20cm_ND_4_7_BayerRG12_Focal_Yes_2.raw	12,747,264	42da7a00
98	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_27cm_ND_4_7_BayerRG12_Focal_No_0.raw	12,747,264	251585ce
99	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_27cm_ND_4_7_BayerRG12_Focal_No_1.raw	12,747,264	ac48d86f
100	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_27cm_ND_4_7_BayerRG12_Focal_No_2.raw	12,747,264	341e092d
101	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_5cm_ND_4_7_BayerRG12_Focal_Yes_0.raw	12,747,264	f7dda135

ID	Domain	Archive path	Bytes	CRC32
102	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_5cm_ND_4_7_BayerRG12_Focal_Yes_1.raw	12,747,264	36ad8f08
103	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_5cm_ND_4_7_BayerRG12_Focal_Yes_2.raw	12,747,264	01a6c736
104	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_8cm_ND_4_7_BayerRG12_Focal_Yes_0.raw	12,747,264	73828437
105	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_8cm_ND_4_7_BayerRG12_Focal_Yes_1.raw	12,747,264	23a9e705
106	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_8cm_ND_4_7_BayerRG12_Focal_Yes_2.raw	12,747,264	a50ab610
107	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_9cm_ND_4_7_BayerRG12_Focal_Yes_0.raw	12,747,264	922b02d2
108	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_9cm_ND_4_7_BayerRG12_Focal_Yes_1.raw	12,747,264	d9e1689f
109	Spatial	532 and 1064/532nm_Spatial/Beam/Exposure_8us_Position_9cm_ND_4_7_BayerRG12_Focal_Yes_2.raw	12,747,264	6cd48e56
110	Spectral (532/1064 nm)	532 and 1064/532nm_Spectrum/Laser_532nm_CW_Acq_1s_Slit_0um_Grad_300.txt	18,089	32d4e987
111	Spatial	532 and 1064/532nm_YesFocal_Gain0_Exposure10k_BayerRG12/Background_Reference.png	7,501,574	1cd026a4
112	Spatial	532 and 1064/532nm_YesFocal_Gain0_Exposure10k_BayerRG12/Position_110mm.png	8,351,433	d426795a
113	Spatial	532 and 1064/532nm_YesFocal_Gain0_Exposure10k_BayerRG12/Position_130mm.png	8,393,219	d668506f
114	Spatial	532 and	8,336,428	216172ce

ID	Domain	Archive path	Bytes	CRC32
		1064/532nm_YesFocal_Gain0_Exposure10k_BayerRG12/Position_150mm.png		
115	Spatial	532 and 1064/532nm_YesFocal_Gain0_Exposure10k_BayerRG12/Position_170mm.png	8,314,679	bc2a7537
116	Spatial	532 and 1064/532nm_YesFocal_Gain0_Exposure10k_BayerRG12/Position_200mm.png	8,271,293	a3afbb33
117	Spatial	532 and 1064/532nm_YesFocal_Gain0_Exposure10k_BayerRG12/Position_250mm.png	8,414,233	80f0bf77
118	Spatial	532 and 1064/532nm_YesFocal_Gain0_Exposure10k_BayerRG12/Position_70mm.png	8,307,963	f2039d69
119	Spatial	532 and 1064/532nm_YesFocal_Gain0_Exposure10k_BayerRG12/Position_90mm.png	8,255,881	ec163f52

Appendix B. Numerical-output catalog

Table B1. Machine-readable analysis outputs

Output file	Rows	Cols	Contents
average_power_inputs.csv	5	11	source | mode | wavelength_nm | repetition_rate_Hz | power_relation | power_value | power_unit | measurement_location | attenuation_state | meter | observed_range
measurement_confidence_matrix.csv	20	4	quantity | reported_result | confidence | basis_and_limitation
power_energy_summary.csv	5	15	source | wavelength_nm | mode | repetition_rate_Hz | reported_power | bound_type | power_low_W | power_high_W | representative_power_W | pulse_energy_low_J | pulse_energy_high_J |
source_data_inventory.csv	119	10	source_id | domain | archive_path | file_name | extension | uncompressed_bytes | compressed_bytes | crc32 | sha256_from_source_manifest | analysis_status

The project ZIP contains every table listed above. The executed notebooks display the full tables without the page-width constraints of this report.

Appendix B.1 Full power-energy table

Table B2. Full power and pulse-energy data

source	wavelength_nm	mode	repetition_rate_Hz	reported_power	bound_type	power_low_W	power_high_W	pulse_energy_low_J	pulse_energy_high_J	photons_per_pulse_low	photons_per_pulse_high
Coherent COMPASS 315M-100	532.00	CW		>130 mW	lower bound	0.13000					
Thorlabs GSL106A	1064.0	CW		>16 mW	lower bound	0.016000					
Thorlabs GSL106A	1064.0	ps	2.0000e+05	>1.35 uW	lower bound	1.3500e-06		6.7500e-12		3.6155e+07	
Thorlabs GSL106A	1064.0	1 ns	2.0000e+05	>9.4 uW	lower bound	9.4000e-06		4.7000e-11		2.5175e+08	
Thorlabs GSL106A	1064.0	65 ns	2.0000e+05	~333 uW (observed 330-340 uW)	observed approximate range	3.3000e-04	3.4000e-04	1.6500e-09	1.7000e-09	8.8379e+09	9.1057e+09

Appendix B.2 Complete confidence matrix

Table B3. Complete measurement-confidence matrix

quantity	reported_result	confidence	basis_and_limitation
532 nm optical power	>130 mW	Moderate	Direct PM100D/S121C reading reported as a strict lower bound; no calibration certificate or repeat series supplied.
532 nm sensor-plane centroid and beam dimensions	Per-frame values in spatial tables	Moderate	Direct camera data and independent repeats; limited by saturation, no dedicated dark frame, background structure, and multi-spot reflections.
532 nm beam ellipticity/orientation	Per-frame values in spatial tables	Moderate	Meaningful for the selected highest-intensity lobe; sensitive to aperture and background.
532 nm M2, major axis	2.539 provisional	Low / provisional	Caustic fit exists, but saturation near waist, limited valid z coverage, unknown z uncertainty, and secondary structure prevent a standards-grade result.
532 nm M2, minor axis	1.814 provisional	Low / provisional	Same limitations as major axis.
1064 nm repetition rate	Approximately 200.0003 kHz	High for measurement chain	Long records and hysteretic trigger-edge detection

quantity	reported_result	confidence	basis_and_limitation
			agree across modes; this is trigger-chain repetition rate, not a standalone intrinsic-jitter certification.
ps electrical response FWHM	20.8896 ns	High as electrical width	The exported waveform supports this electrical response width; it is not the optical pulse duration.
1 ns electrical response FWHM	32.3680 ns	High as electrical width	The detector/electronics do not resolve a 1 ns optical pulse.
65 ns electrical response FWHM	49.0250 ns	Moderate as electrical width	Partially resolved but detector-convolved and affected by ringing; not a calibrated optical duration.
ps pulse energy	>6.75 pJ	Moderate lower bound	Derived exactly from the reported lower-bound average power and measured/nominal 200 kHz rate.
1 ns pulse energy	>47.0 pJ	Moderate lower bound	Derived exactly from the reported lower-bound average power and 200 kHz rate.
65 ns pulse energy	1.65-1.70 nJ	Moderate approximate range	Derived from the reported 330-340 uW range at 200 kHz.
Peak power in any pulsed mode	Not determined	Not supported	Requires trustworthy optical pulse shape and duration; detector traces are insufficient.
532 nm spectral center	Near 532 nm; see spectral tables	Moderate	Visible Hg references bracket the green wavelength, but repeatability and configuration-specific calibration still govern accuracy.
532 nm intrinsic linewidth	Not determined	Not supported	Narrow features are undersampled and instrument limited.
1064 nm narrow spectral center	Approximately 1059-1061 nm measured	Low absolute confidence	No near-IR reference; visible offset calibration is extrapolated far beyond 546.1 nm.
1064 nm narrow intrinsic linewidth	Not determined	Not supported	Strong slit dependence and insufficient sampling demonstrate instrument limitation.
ps spectral FWHM	8.492-9.606 nm	Moderate to high	Broad feature is well sampled; width is more robust than absolute center wavelength.
ps transform-limit lower bound	0.170-0.192 ps Gaussian; 0.122-0.137 ps sech^2	Conditional	Calculated from spectral intensity width assuming a pulse shape and zero spectral phase; not a measured temporal duration.
Absolute wavelength accuracy near 1064 nm	Unverified	Not established	Requires a traceable near-IR line/reference measured with the same grating and slit configuration.

Appendix C. Definitions and equations

Table C1. Principal equations

Quantity	Equation	Use in this report
Pulse energy	E_pulse = P_average / f_rep	Requires average power and repetition rate only.
Photon energy	E_photon = h c / lambda	Photon counts use nominal 532 or 1064 nm.
Photon number per pulse	N = E_pulse / E_photon	Inherits power-bound/range semantics.
Second-moment diameter	D4sigma = 4 sqrt(<(x-<x>)^2>)	Evaluated on principal axes after background correction and selected-lobe aperture.
Gaussian beam caustic	w(z)^2 = w0^2 + [M2 lambda (z-z0)/(pi w0)]^2	Used only for provisional caustic sensitivity analysis.
Frequency bandwidth	Delta nu approximately c Delta lambda / lambda0^2	Small-fractional-bandwidth conversion; numerical notebook uses the sampled wavelength axis.
Transform limit	Delta t >= TBP / Delta nu	TBP depends on assumed intensity shape; zero spectral phase gives only a lower bound.

Constants used: h = 6.62607015 x 10^-34 J s and c = 299792458 m/s. Numerical values are reported with at least three significant figures where the underlying data justify them. Extra displayed digits in repeated or fitted values support traceability and should not be interpreted as full absolute accuracy.

Appendix D. Reproducibility and manufacturer documentation

D.1 Project contents

- Five original domain notebooks plus the new complete report/data-audit notebook, each with clean and executed versions where applicable.
- Reusable laser_characterization.py analysis module and experiment_metadata.json.
- Complete source inventory, source/data manifests, output catalog, traceability map, confidence matrix, and power-energy table.
- Publication-resolution figures and all machine-readable CSV/JSON outputs.
- DOCX and PDF versions of this report.

D.2 Reproduction

Place the original “532 and 1064” archive or extracted folder under project/data, install the packages listed in requirements.txt, open 00_Start_Here.ipynb, and run the notebooks in numerical order. The data-audit notebook can be run after the domain notebooks. The source archive hash and per-file manifest should be checked before interpreting differences between reruns.

D.3 Manufacturer documentation

- Basler, acA3088-57uc official camera documentation: sensor format, 3088 x 2064 resolution, 2.4 um pixel pitch, and supported pixel formats.
- Coherent, COMPASS 315M series legacy product documentation.
- Thorlabs, GSL106A gain-switched laser product documentation.
- Thorlabs, PDA36A legacy amplified silicon detector documentation.
- Thorlabs, PM100D power meter, S121C sensor, and F280APC-1064 collimator documentation.
- HORIBA, iHR550/iHR series spectrometer and Synapse Plus detector documentation; LabSpec 6 acquisition documentation.
- Keysight/Agilent, DSO6104A oscilloscope and 33120A function generator documentation.

Manufacturer specifications describe nominal instrument capability. They are not used as substitutes for source-specific calibration, a measured instrument response, or a complete uncertainty budget.

Appendix E. Complete figure gallery

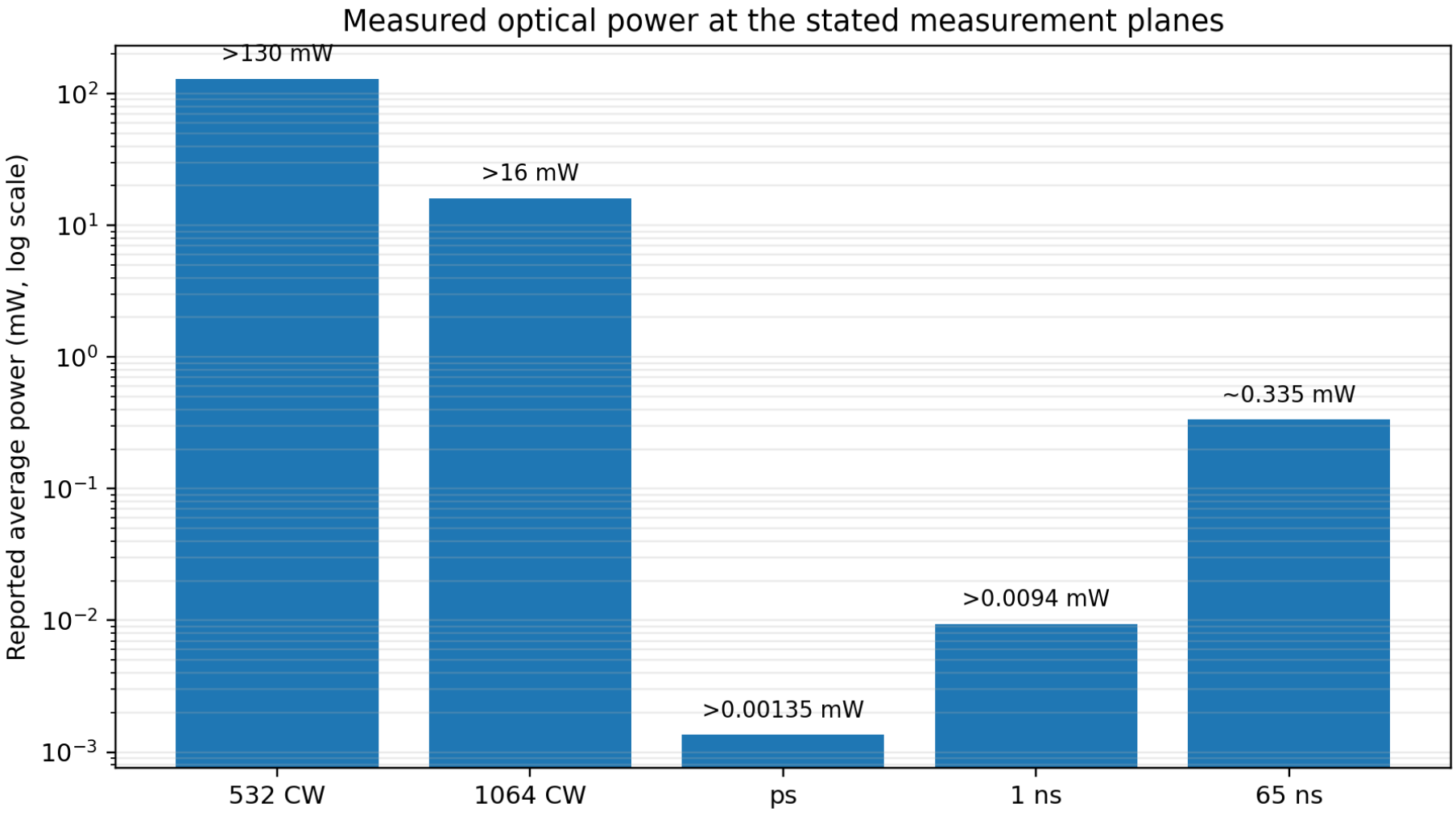


Figure E1. power measurements v3.

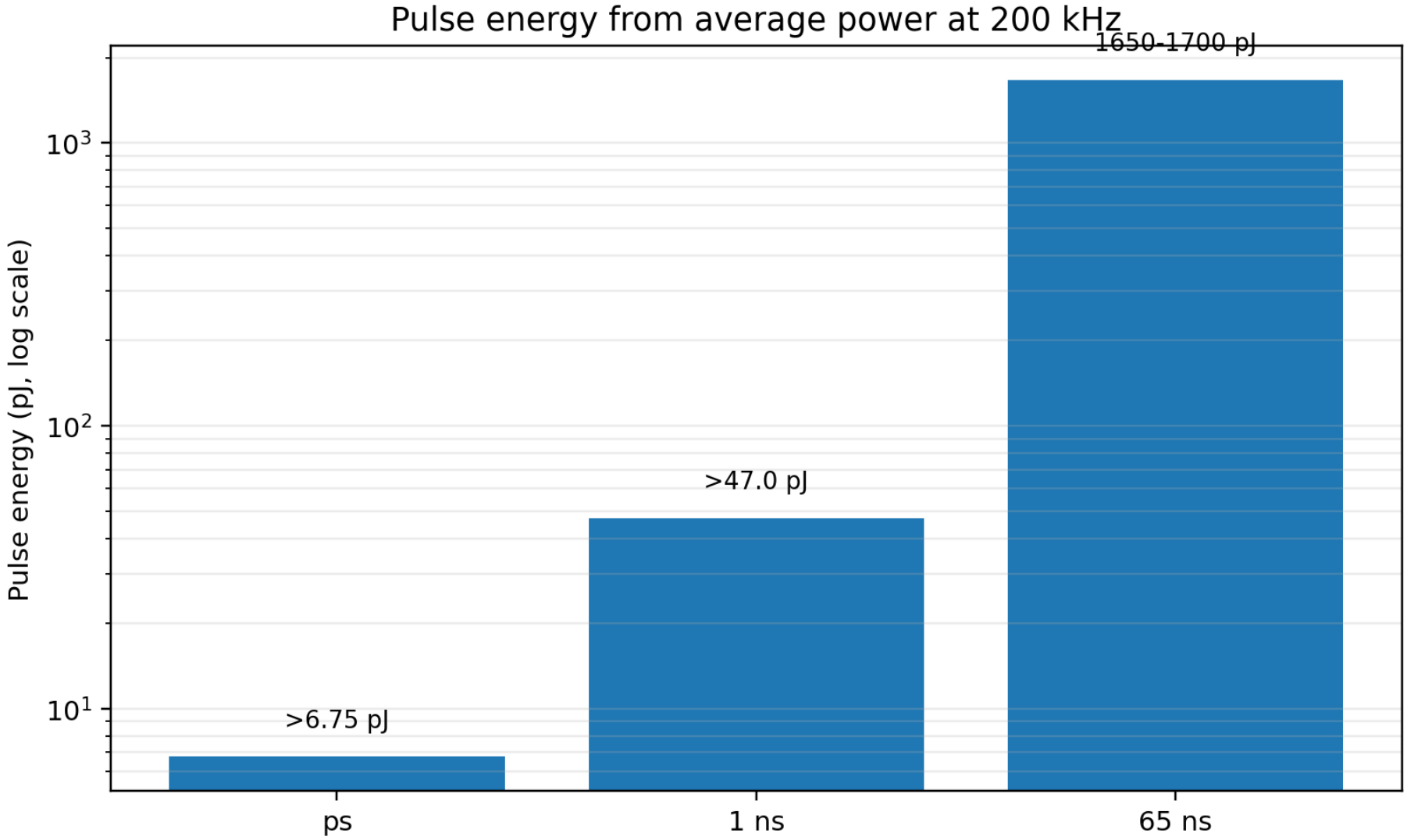


Figure E2. pulse energy v3.